Mathematics - K-12 Level Assessment 20 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. During a semester, a student received scores of 76, 80, 83, 71, 80, and 78 on six tests. What is the student's average score for these six tests?
   (a) 76
   (b) 77
   (c) 78
   (d) 79

2. Solve 18.5 dol over m gal = 3.60 dol over 7.5 gal. Round to the nearest hundredth if necessary.
   (a) 8.86
   (b) 42.25
   (c) 32.54
   (d) 38.44

3. What is the smallest positive integer with factors of 16, 15, and 12?
   (a) 12
   (b) 22
   (c) 840
   (d) 240

4. What is the shortest distance from the origin to the circle defined by $x^2 - 24x + y^2 + 10y + 160 = 0$?
   (a) 10
   (b) 16
   (c) 24
   (d) 12

5. Evaluate -2(x - 3) for x = 2.
   (a) -4
   (b) -2
   (c) 10
   (d) 2

True / False (10 marks)

Answer True or False

6. True or False: Jane’s mean quiz score is 94.5, which is the result of adding her scores and dividing by five.

7. True or False: The ratio of x to y is 5 to 4 when the ratio of 2 x - y to x + y is 2 to 3.

8. True or False: The interval consisting of all u such that neither 2 u nor -20 u is in (-, -1) is [-1, 1].

9. True or False: The correct answer to ’Evaluate $\log_3 81$.’ is ’-1’.

10. True or False: You can buy 12 cans of pears for $9.48if3canscost2.37$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. An ice cream shop sold 48 vanilla milkshakes, which represented 40% of its total sales for the day. Calculate the total number of milkshakes sold and explain your reasoning.

12. A baker made 232 muffins and sent 190 to a hotel. Describe how you would determine the number of boxes needed for the remaining muffins, including the relevant equation.

End of Paper